

Contribution Report

Ensemble Methods: Boosting vs. Bagging
AI Academy – Machine Learning Final Project, Spring 2026

Team Sigmoid
July 2026

Team Members

Full Name	GitHub Username	Email
Emin Huseynli	ehuseynli96-ops	emin.huseynli25@aiacademy.az
Ziyad Muradov	thedarkstringg	ziyad.muradov25@aiacademy.az
Ismayil Yusifli	ismayilysfli	ismayil.yusifli25@aiacademy.az

Individual Contributions

Member	Modules / Code	Report Sections	Experiments
Emin Huseynli ehuseynli96-ops	src/trees/decision_tree.py src/boosting/adaboost.py src/boosting/samme_r.py experiments/run_all.py experiments/bonus_samme_r.py	§2 Decision Tree §3 AdaBoost	Exp. 1, Exp. 2
Ziyad Muradov thedarkstringg	src/bagging/random_forest.py src/boosting/one_vs_rest_adaboost.py src/utils/preprocessing.py src/utils/datasets.py src/metrics/evaluation.py	§4 Random Forest §6 Experimental Setup	Exp. 3, Exp. 4, Exp. 5
Ismayil Yusifli ismayilysfli	src/unsupervised/pca.py src/unsupervised/kmeans.py src/unsupervised/dbscan.py experiments/*.py (1-7) experiments/bonus_tsne.py experiments/common.py	§1 Introduction §7 Unsupervised Conclusion	Exp. 6, Exp. 7

Sign-off

All three members reviewed and approved the final codebase, report, and experimental results via GitHub pull request review.

By submitting this report, each undersigned member confirms that the contributions listed above accurately reflect their individual work on this project.

Emin Huseynli

Emin Huseynli
July 15, 2026

Ziyad Muradov

Ziyad Muradov
July 15, 2026

Ismayil Yusifli

Ismayil Yusifli
July 15, 2026